

Hiring Mgr.: Soumya_dey@keysight.com

Job Title: Software Engineering Intern – Thermal Numerical Computation Engr.

Overview: This intern position is for the Thermal Simulation Squad in Keysight EDA division. We are seeking a motivated Software Engineering Intern (with background in numerical computation) to join our EDA Thermal Simulation software team. The intern will work on tools and technologies to:

- Enhance and maintain an IC electro-thermal simulator and its interfaces that are integrated into Keysight ADS and Cadence Virtuoso.
- Support graphical interfaces for smooth, surface and volume plots, and their underlying computations.
- Support co-simulation with circuit simulators in ADS, and with other tools for power computation.

This role is ideal for candidates with strong C++ programming skills, a solid mathematical foundation, and a keen interest in learning complex software systems used in electronic design automation.

Project Title 1, 2 & 3:

- Fix and augment Keysight HeatWave (Electrothermal Simulation) GUI, including artifacts remaining after recent update
- Display X and Y plane views of layout with smooth temperature plots or contours overlaid
- Prototype co-sim with other solvers, especially data preparation that could be leveraged in production code.

Required Qualifications & Experience

- A few years coding in Modern C++, using STL and Boost, in Linux; working knowledge of python/numpy/scipy/matplotlib etc.
- Some EE background including circuit design and some understanding of VLSI layout
- Some understanding of physical-design and related computational methods, including boolean operations on layout.

- Some knowledge of numerical methods related to solving PDEs.
- Experience with scripting languages such as Tcl
- Some understanding of build related utilities: Cmake, make, conan etc.

What the Intern Will Learn

- Practical application of C++ in large-scale engineering software.
- Use of geometry and coordinate-based algorithms in real-world thermal simulation tools.
- Application of numerical methods
- During problem solving learn about Floorplanner, therefore will learn how to:
 - Display X and Y plane views of layout with smooth temperature plots or contours overlaid
 - Prototype co-sim with other solvers, especially data-preparation that could be leveraged in production code.